

Anselm Paulus

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✂ [AnselmPaulus](#) • [a-paulus](#) • Last updated on May 27, 2026

Current Position

Ph.D. Student in Machine Learning, *University of Tuebingen & Max-Planck-Institute for Intelligent Systems*, Tuebingen, Germany (supervised by [Georg Martius](#) on differentiable optimization and LLM safety/security) 2022 – 2026

Education

M.Sc. in Computer Science, *University of Tuebingen* (4.00/4.00) 2021 – 2022

B.S. in Computer Science, *University of Tuebingen* (3.73/4.00) 2018 – 2021

B.S. in Physics, *University of Tuebingen* (3.80/4.00) 2015 – 2021

Previous Positions

Research Scientist Intern, *Meta, Fundamental AI Research (FAIR)*, Menlo Park, CA (with [Arman Zharmagambetov](#) on game-theoretical perspectives on LLM safety) 2025 – 2026

Research Scientist Intern, *Meta, Fundamental AI Research (FAIR)*, New York City (with [Brandon Amos](#) on amortized optimization for LLM safety) 2023 – 2024

Research Assistant, *Max-Planck-Institute for Intelligent Systems*, Tuebingen, Germany (with [Georg Martius](#) on differentiable combinatorial optimization) 2018 – 2022

Publications [Google Scholar: 1.1k+ citations]

2026.....

1. *Hard Contacts with Soft Gradients: Refining Differentiable Simulators for Learning and Control*
Anselm Paulus*, [Andreas René Geist*](#), [Pierre Schumacher](#), [Vít Musil](#), and [Georg Martius](#)
ICLR 2026
2. *Safety Alignment of LMs via Non-cooperative Games*
Anselm Paulus, [Ilia Kulikov](#), [Brandon Amos](#), [Rémi Munos](#), [Ivan Evtimov](#), [Kamalika Chaudhuri](#), and [Arman Zharmagambetov](#)
ICML 2026 (Spotlight)
3. *SoftJAX & SoftTorch: Empowering Automatic Differentiation Libraries with Informative Gradients*
Anselm Paulus, [A. René Geist](#), [Vít Musil](#), [Sebastian Hoffmann](#), [Onur Beker](#), and [Georg Martius](#)
ICML 2026 (Oral)

2025.....

4. *AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs* [code]
Anselm Paulus*, [Arman Zharmagambetov*](#), [Chuan Guo](#), [Brandon Amos†](#), and [Yuandong Tian†](#)
ICML 2025

2024.....

5. *LPGD: A General Framework for Backpropagation through Embedded Optimization Layers* [code]
Anselm Paulus, [Georg Martius](#), and [Vít Musil](#)
ICML 2024

2023.....

6. [Backpropagation through Combinatorial Algorithms: Identity with Projection Works](#) [code]
Anselm Paulus*, Subham Sekhar Sahoo*, Marin Vlastelica, Vít Musil, Volodymyr Kuleshov, and Georg Martius
ICLR 2023

2021.....

7. [CombOptNet: Fit the Right NP-Hard Problem by Learning Integer Programming Constraints](#) [code]
Anselm Paulus, Michal Rolínek, Vít Musil, Brandon Amos, and Georg Martius
ICML 2021 (Spotlight, Oral at LMCA NeurIPS 2020 workshop)

2020.....

8. [Differentiation of Blackbox Combinatorial Solvers](#) [code]
Anselm Paulus*, Marin Vlastelica P.*, Vít Musil, Georg Martius, and Michal Rolínek
ICLR 2020 (Spotlight)
9. [Optimizing Rank-based Metrics with Blackbox Differentiation](#) [code]
Michal Rolínek*, Vít Musil*, **Anselm Paulus**, Marin Vlastelica P., Claudio Michaelis, and Georg Martius
CVPR 2020 (Oral, best paper award nomination)
10. [Deep Graph Matching via Blackbox Differentiation of Combinatorial Solvers](#) [code]
Michal Rolínek, Paul Swoboda, Dominik Zietlow, **Anselm Paulus**, Vít Musil, and Georg Martius
ECCV 2020

Open Source Repositories

0.8k+ GitHub stars across all repositories.

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|---|------|
| 1. a-paulus/softjax — ★74 — <i>SoftJAX (library for differentiable operations in JAX)</i> | 2026 |
| 2. a-paulus/softtorch — ★72 — <i>SoftTorch (library for differentiable operations in PyTorch)</i> | 2026 |
| 3. facebookresearch/advgame — ★20 — <i>AdvGame (Adversarial attacks on LLMs)</i> | 2026 |
| 4. facebookresearch/advprompter — ★181 — <i>AdvPrompter (Adversarial attacks on LLMs)</i> | 2025 |
| 5. martius-lab/diffcp-lpgd — ★3 — <i>Lagrangian Proximal Gradient Descent (now merged into CVXPY)</i> | 2024 |
| 6. martius-lab/solver-differentiation-identity — ★12 — <i>Blackbox Identity Differentiation</i> | 2023 |
| 7. martius-lab/CombOptNet — ★74 — <i>CombOptNet</i> | 2021 |
| 8. martius-lab/blackbox-deep-graph-matching — ★89 — <i>Deep Graph Matching</i> | 2020 |
| 9. martius-lab/blackbox-backprop — ★355 — <i>Blackbox Differentiation</i> | 2020 |

Professional Activities

Invited Talks.....

1. *AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs* — Masaryk University, Brno, Czechia 2024
2. *On differentiable combinatorial optimization* — [Dagstuhl Seminar: Machine Learning and Logical Reasoning: The New Frontier](#) 2022

Reviewing.....

International Conference on Learning Representations (ICLR)	2022, 2023, 2024, 2025, 2026
International Conference on Machine Learning (ICML)	2022, 2023, 2024, 2025
Neural Information Processing Systems (NeurIPS)	2021, 2022, 2025
Neural Information Processing Systems (NeurIPS) DiffCoAlg Workshop	2025